

## Answer Key — General Science Grade 10

Sub-Strand 1.3: Nutrition in Animals • Lesson 1: The Ugali and Sukuma Wiki Puzzle — Launching the Anchoring Phenomenon

**Teacher reference only. Do not project or hand out.**

### Quick marking grid

| Q1 | Q2 | Q3 | Q4 | Q5 | Q6 | Q7 | Q8 |
|----|----|----|----|----|----|----|----|
| B  | A  | B  | C  | B  | C  | B  | B  |

USE AFTER: PREDICT PHASE — Predict: individual Prediction Sheet (UGALI, SUKUMA WIKI, NYAMA CHOMA, WHAT I AM NOT SURE ABOUT)

**Q1. In the Predict activity, each learner fills in a Prediction Sheet about a lunch of ugali, sukuma wiki and nyama choma. Which is the best description of what you should write?**

- A. One prediction only, about the food you like most
- B. What you honestly think happens to each food after it is swallowed, including what you are not sure about**
- C. Only the facts the teacher has already taught, so that nothing is wrong
- D. A copy of your partner's prediction, so that the class agrees

**Correct answer: B.** The Predict phase asks learners to write their own current thinking for each food, with a column for what they are unsure about. Predictions are not corrected and there are no wrong answers at this stage.

USE AFTER: OBSERVE PHASE — Observe: 'Slow Look' and the NOTICE card

**Q2. During the 'Slow Look', learners must write only what they can observe on the NOTICE side of a card, without explaining. Which statement is an observation and not an explanation?**

- A. The ugali is white, smooth and dense.**
- B. The nyama choma gives us protein.
- C. The sukuma wiki is healthy for the body.
- D. The ugali contains carbohydrates.

**Correct answer: A.** Colour, texture and structure can be seen directly, so this is an observation. The other statements are claims about nutrients or health that go beyond what is seen.

USE AFTER: OBSERVE PHASE — Observe: three anomaly prompts, PUZZLE 3 (CANNOT YET EXPLAIN side of the card)

**Q3. The teacher says: 'Your stomach is made of protein. Your stomach also produces acid and protein-digesting enzymes.' What puzzle does this raise?**

- A. Why does the small intestine absorb nutrients into the blood?
- B. Why does the stomach not digest itself?**
- C. Why does ugali digest faster than nyama choma?
- D. How does the body know which digestive chemical to release, and when?

**Correct answer: B.** A stomach made of protein that also makes protein-digesting chemicals raises the puzzle of why it does not digest itself. This is one of the three anomalies recorded on the CANNOT YET EXPLAIN side of the card.

USE AFTER: EXPLAIN PHASE — Explain: 'What Do We Already Know?' gallery sort (CONFIDENT | THINK SO | NO IDEA)

**Q4. In the class gallery sort, the teacher reads out a prediction and counts the hand signals: 18 thumbs-up (we know this is true), 7 open hands (we think so) and 3 thumbs-down (no idea). Where does the teacher place that prediction?**

- A. THINK SO, because some learners were unsure
- B. It is taken off the board until the end of the unit
- C. CONFIDENT, because most learners showed thumbs-up**
- D. NO IDEA, because some learners had no idea

**Correct answer: C.** The sticky note goes in the column chosen by the majority. 18 of the 28 learners gave thumbs-up, so it goes under CONFIDENT.

USE AFTER: EXPLAIN PHASE — Explain: 'Known Foundation' identified in the gallery sort

**Q5. In the gallery sort, the class looks for prior knowledge about each food that they are confident is true. Which matching of food and main component do learners describe as a visible part of the Kenyan meal?**

- A. Ugali is mainly protein, sukuma wiki is mainly starch, and nyama choma is mainly fibre.
- B. Ugali is mainly starch, sukuma wiki is mainly fibre, and nyama choma is mainly protein.**
- C. Ugali is mainly fibre, sukuma wiki is mainly protein, and nyama choma is mainly starch.
- D. All three foods contain only starch.

**Correct answer: B.** Ugali (maize meal) is starchy, sukuma wiki (kale) is fibrous and nyama choma (grilled meat) is protein-rich. These are the three kinds of food matter in the meal.

USE AFTER: DQB PHASE — DQB: three-colour sticky notes (YELLOW noticed, BLUE wonder, GREEN know)

**Q6. On the Driving Question Board, a learner has a sticky note that says: 'Why does my stomach not digest itself?' Which colour of note does it belong on?**

- A. No note. The teacher answers it straight away
- B. Yellow: something I noticed about the meal
- C. Blue: something I wonder or a question I have**
- D. Green: something I think I already know

**Correct answer: C.** Blue is for wonders and questions. Yellow is for observations and green is for things learners think they already know. Questions are collected, not answered, in this lesson.

USE AFTER: MODEL BUILDING PHASE — Model: first 'Food Journey Map' (DRAFT 1 — Before Investigation)

**Q7. Learners label their first Food Journey Map 'DRAFT 1 — Before Investigation' and are told not to erase anything. Why?**

- A. Draft 1 has to be correct so that it can be used as the final answer
- B. It records their honest current thinking, so they can compare it with later drafts and see how their understanding changes**
- C. Mistakes must be hidden so that the final map looks perfect
- D. The teacher will mark it and give a score for each correct part

**Correct answer: B.** The map is the first version of a model that will be revised in later lessons. Mistakes are treated as data that show where learning grows, so nothing is erased or marked.

USE AFTER: END OF LESSON — End of lesson: after the DQB and Food Journey Map, reflecting on the three puzzles

**Q8. At the end of this lesson, the class has recorded 'Why is ugali digested faster than nyama choma?' as a puzzle. What is the status of this question?**

- A. It was answered by the Driving Question card
- B. It is an open question that has not been explained yet, and the class will investigate it during the unit**
- C. The teacher explained it during the gallery sort
- D. The class voted and agreed on the answer

**Correct answer: B.** This lesson opens the inquiry and does not resolve any of the three puzzles. The question is recorded for investigation in later lessons.